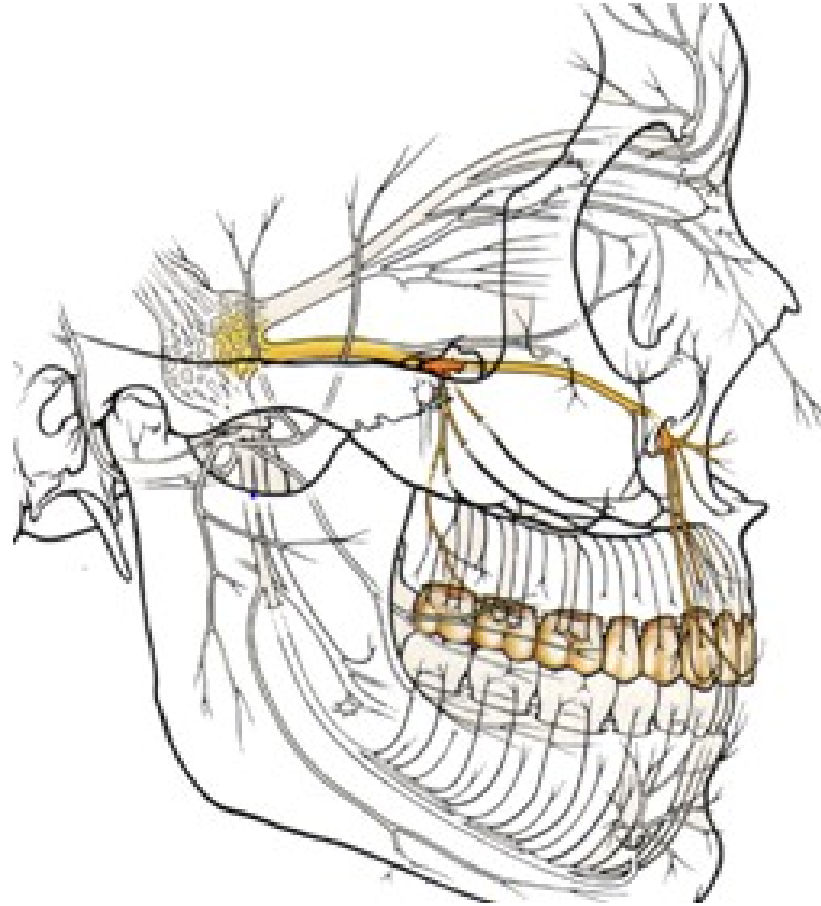


Techniques of maxillary anesthesia

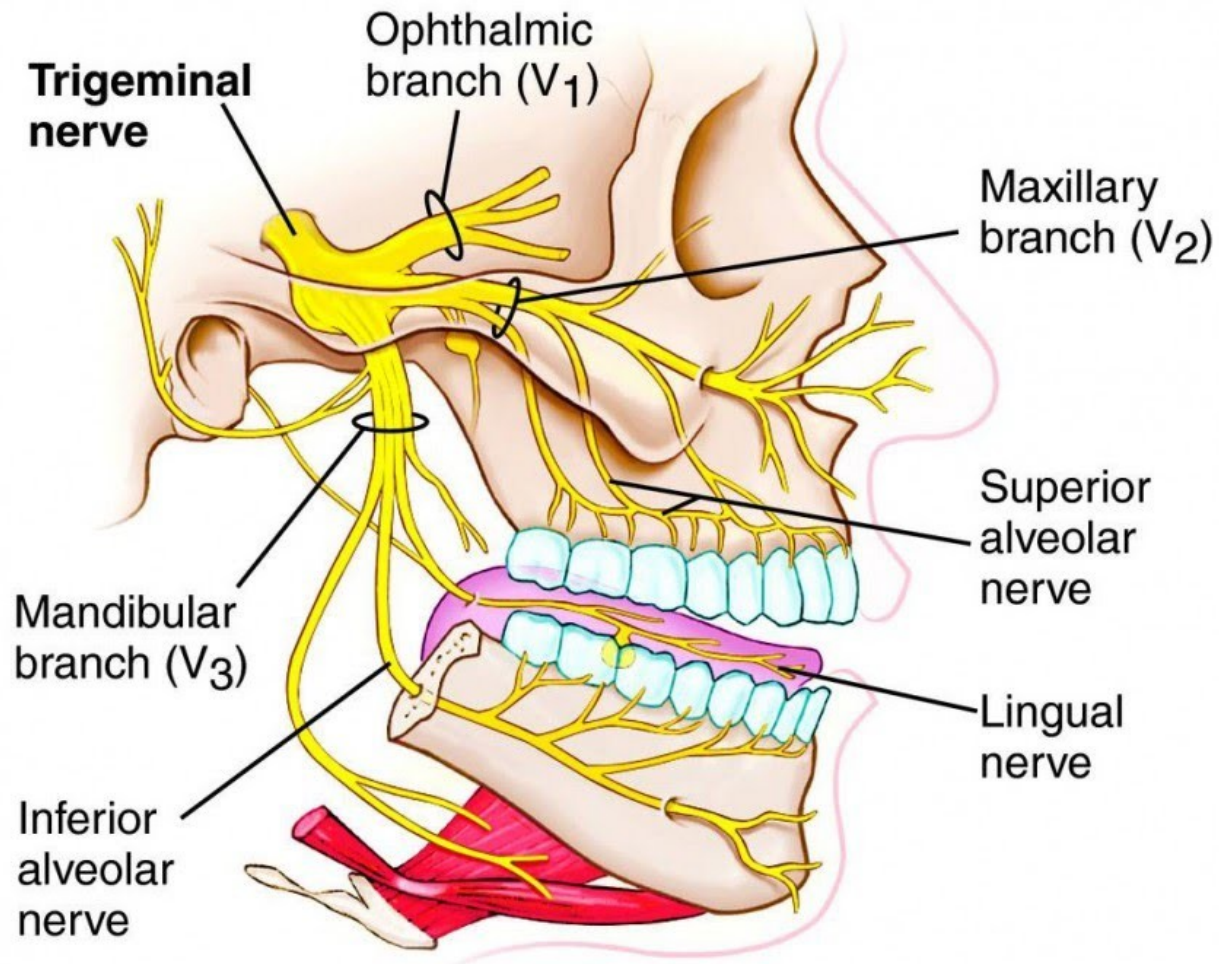


3 BDS
Local Anesthesia Lecture

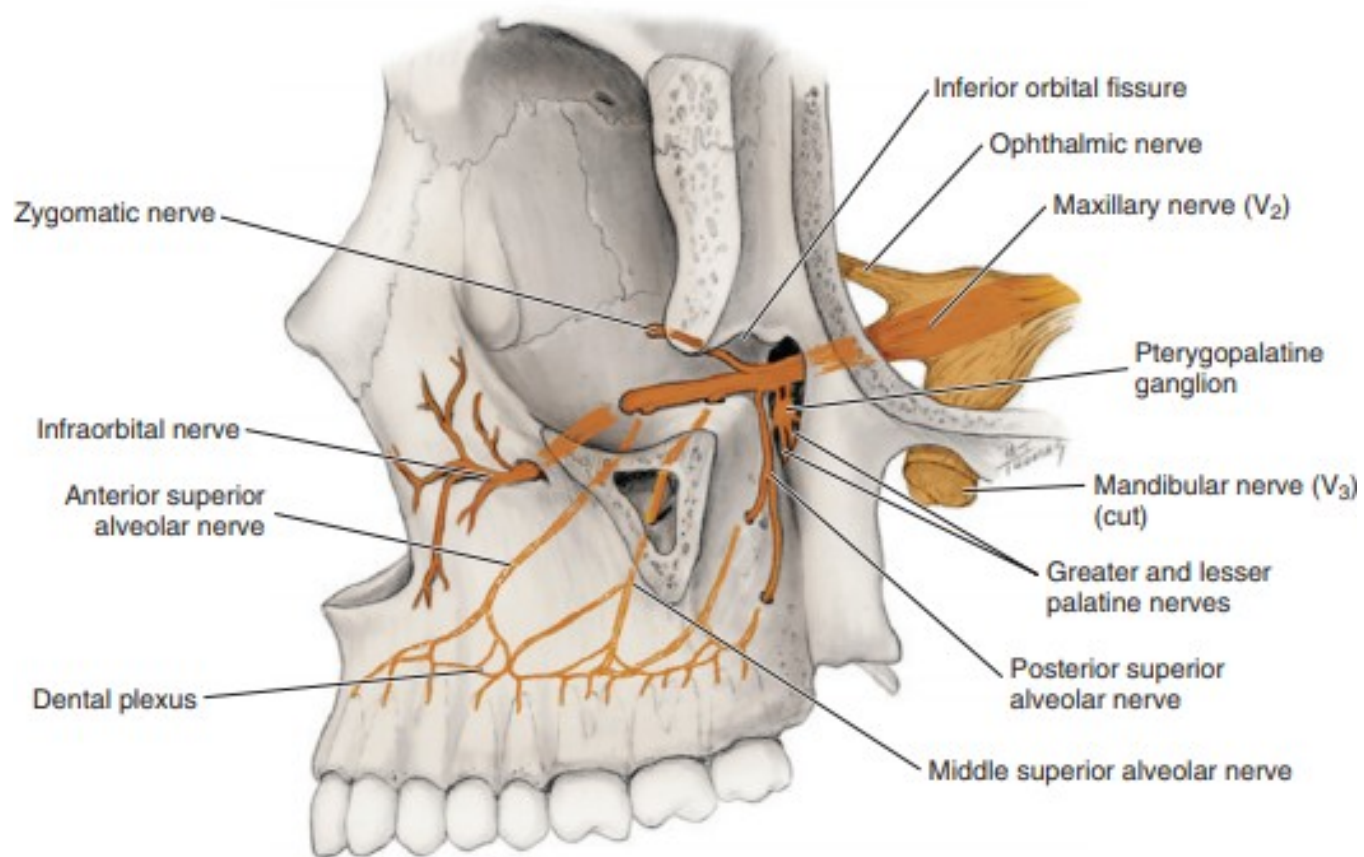
By the end of this lecture you should be able to:

- List the land marks for maxillary anesthetic technique
- Define the different Maxillary Nerve local anesthetic techniques
- Explain the suitability of Maxillary Nerve local anesthetic techniques for maxillary teeth extraction

Trigeminal Nerve

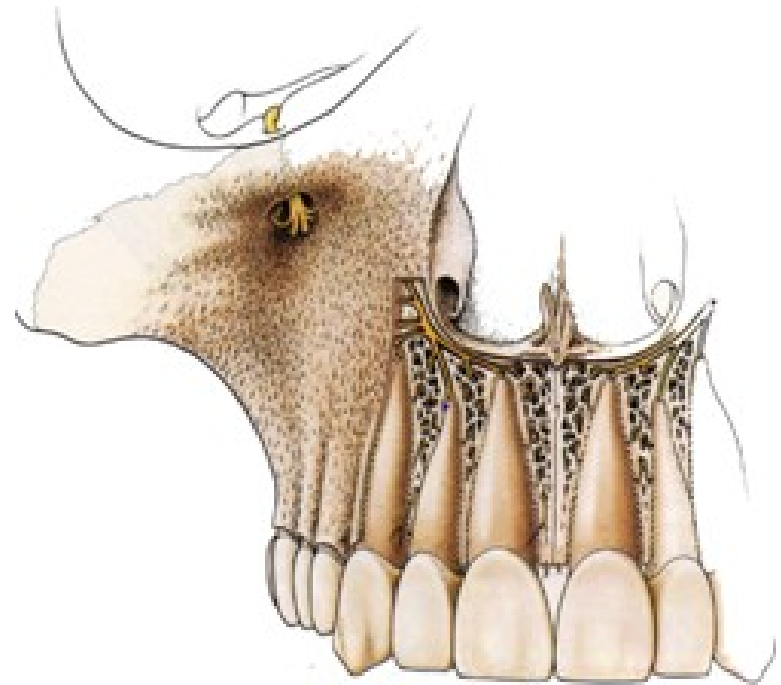


The Maxillary Nerve (V₂)

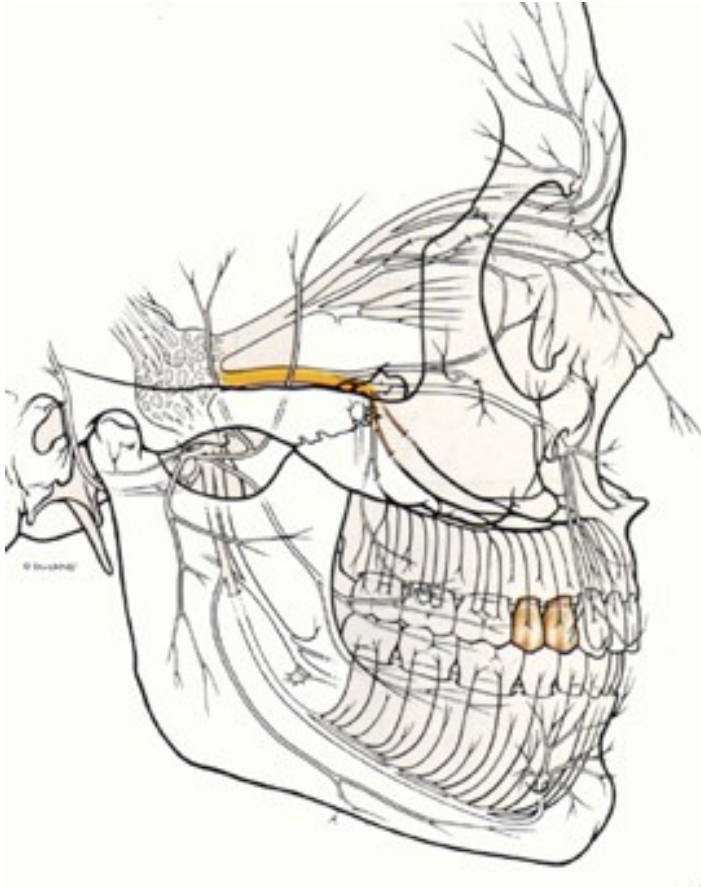


Nerve supply of the Upper Incisors & Canines

- The anterior maxilla is innervated by the **anterior superior alveolar nerves** which branch off the infraorbital canal.
- The maxilla has a thin and porous cortex which is easily penetrated by the LA solution injected at the level of the apex of the teeth

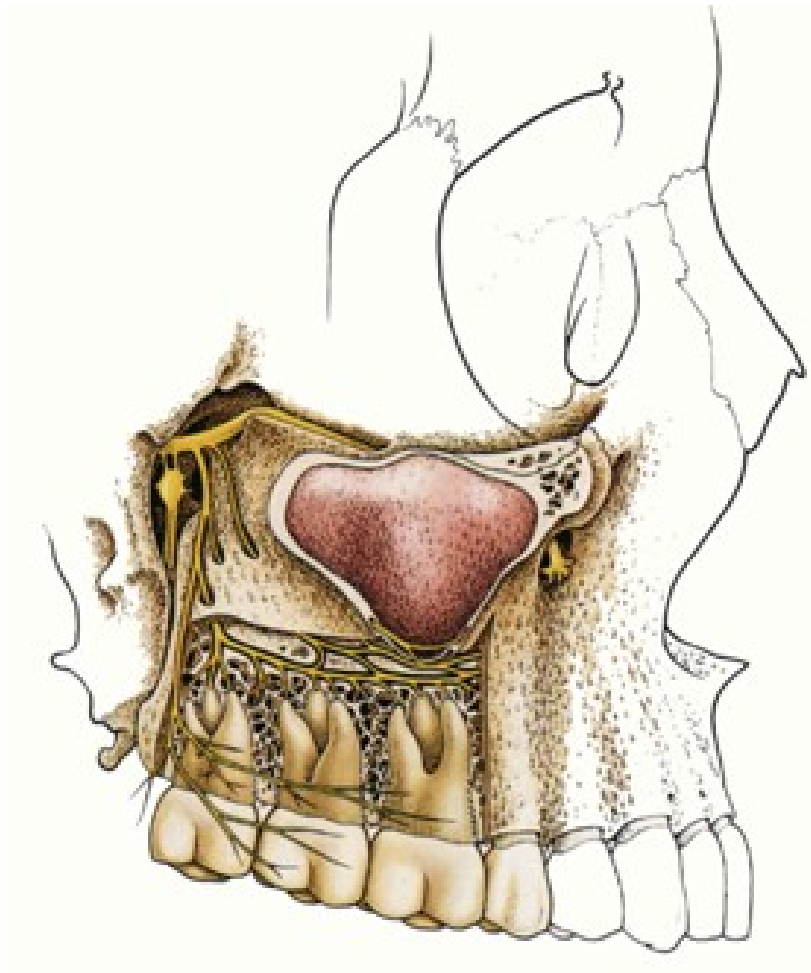


Nerve supply of the Upper Premolars



Middle Superior Alveolar Nerve

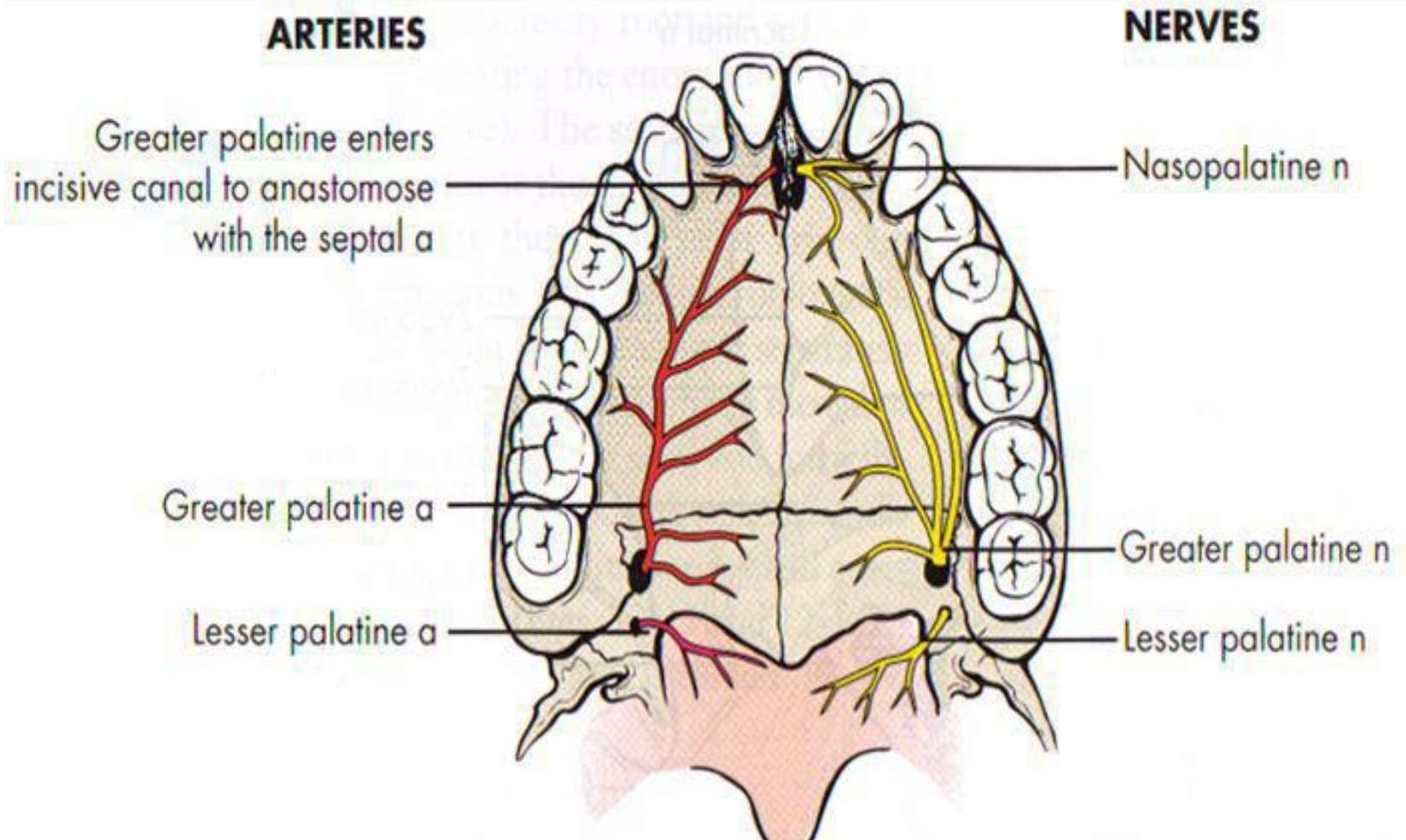
- The nerve is present in 28% of cases.
- When present it innervates
 - upper Premolars
 - Buccal gingiva
 - Periosteum
 - Mesiobuccal root of 1st molar



Upper Molars

- Upper molars are innervated by the **posterior superior alveolar nerve** which is a branch of the infraorbital nerve before it reaches the orbital cavity

The palate



Supraperiosteal Injection

commonly referred to as

infiltration anesthesia

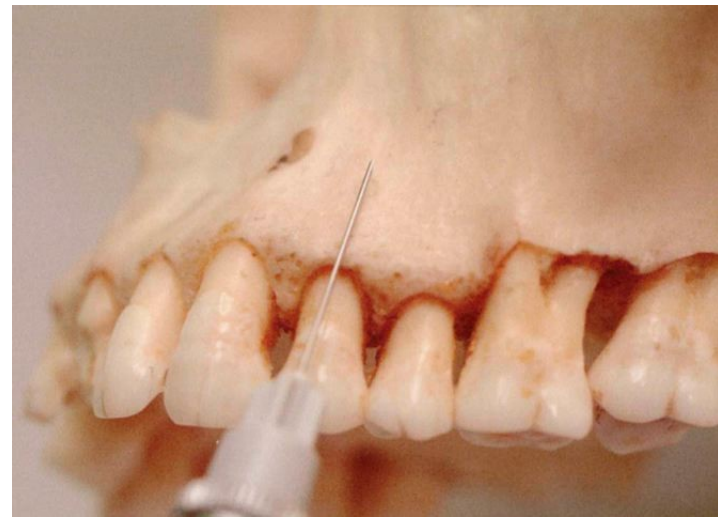
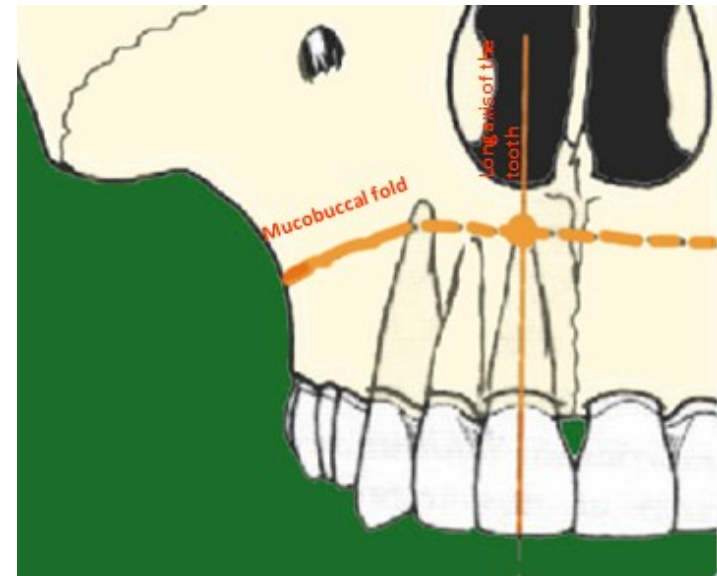
- The supraperiosteal injection can be achieved by depositing the anesthetic solution adjacent to the periosteum overlying the apex of the tooth to be anesthetized.
- In the case of the maxillary arch with its relatively porous bone, the anesthetic solution readily diffuses through its thinner facial plates of the maxillary bone to anesthetize the terminal fibers of the dental plexus



Localization of the Point of Needle Insertion in infiltration technique

Land Marks

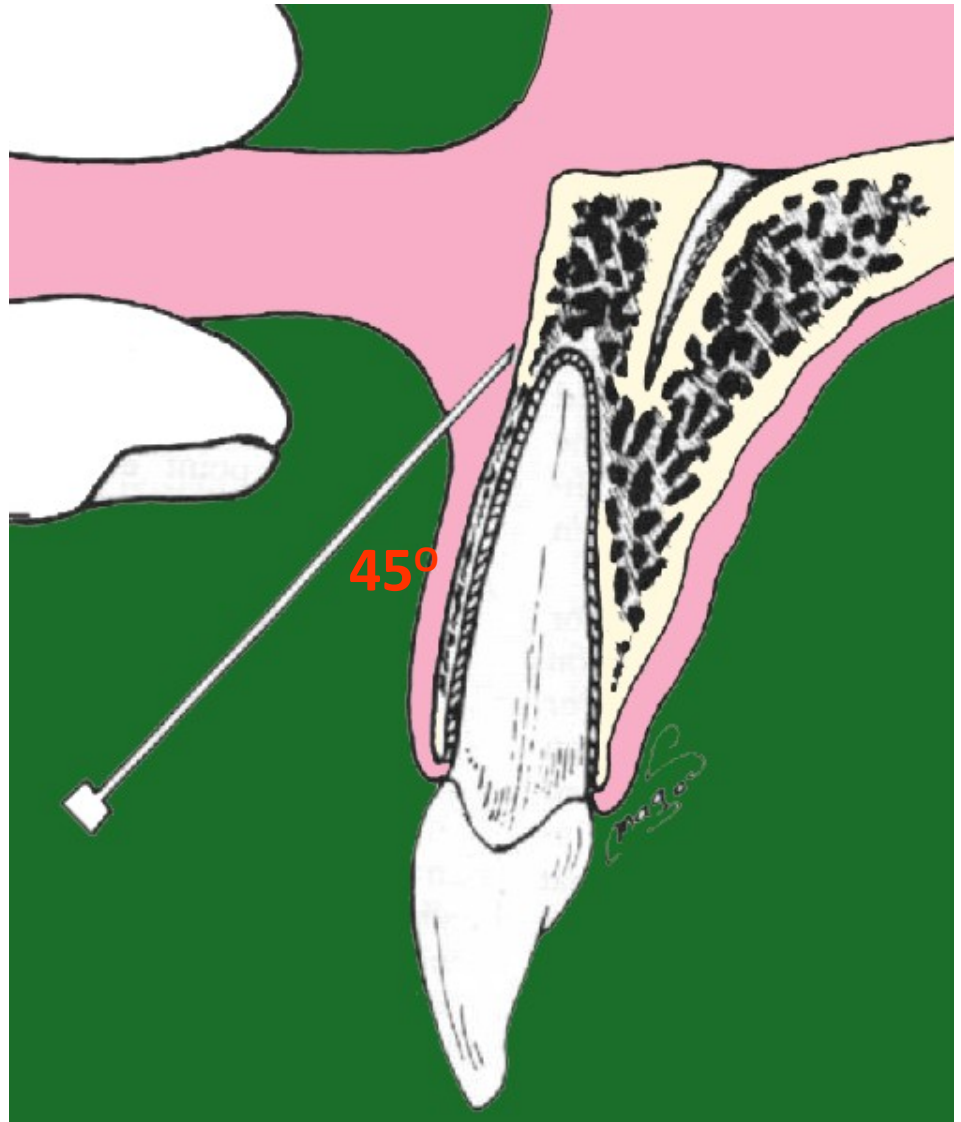
- The point of needle insertion is the point of intersection of two imaginary lines.
- Horizontal line parallel to the mucobuccal fold and a vertical line parallel to the long axis of the target tooth



Needle Insertion on the Labial Infiltration Injection

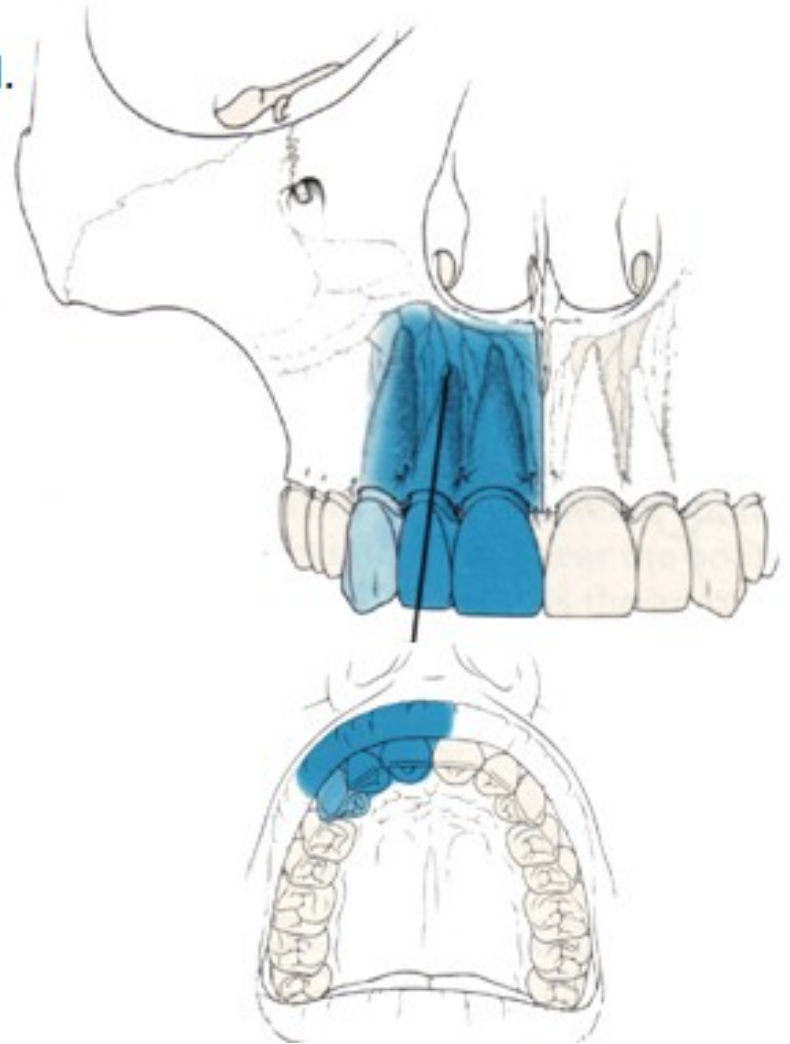
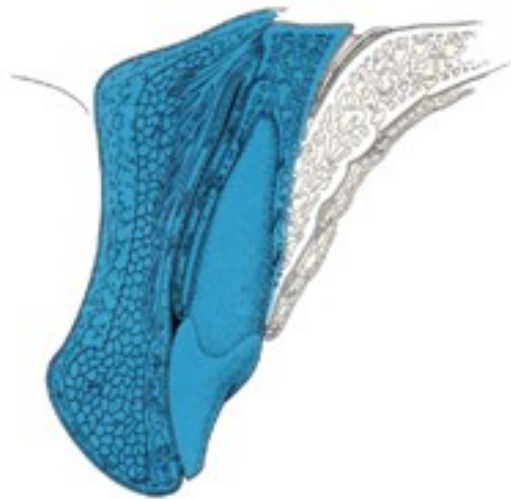
- Lift lip and pull tissue taut
- Hold syringe in a line with the long axis of the target tooth and making **45 degree angle** with the buccal plate of bone

Orient needle bevel to face bone



When in contact with bone a slow injection of 0.6-0.8 ml of solution (one third of cartridge) is performed. Do not allow tissues to baloon

This will affect anesthesia of the target tooth and one or both of the adjacent teeth.



Infiltration Technique



Infiltration anesthesia

- Contra indications
- infection or acute inflammation in the area of injection
- Permanent maxillary first molar in children ,as its apex might be located in dense zycomatic bone

Advantages

- High Success rate
- Technically easy
- Atraumatic

Disadvantages

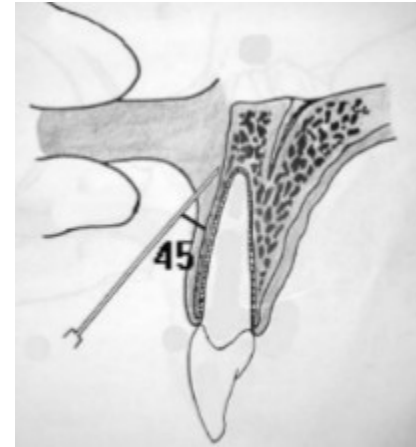
Not recommended for large areas- because need multiple needle insertion & large volume of Anesthesia

Signs and Symptoms

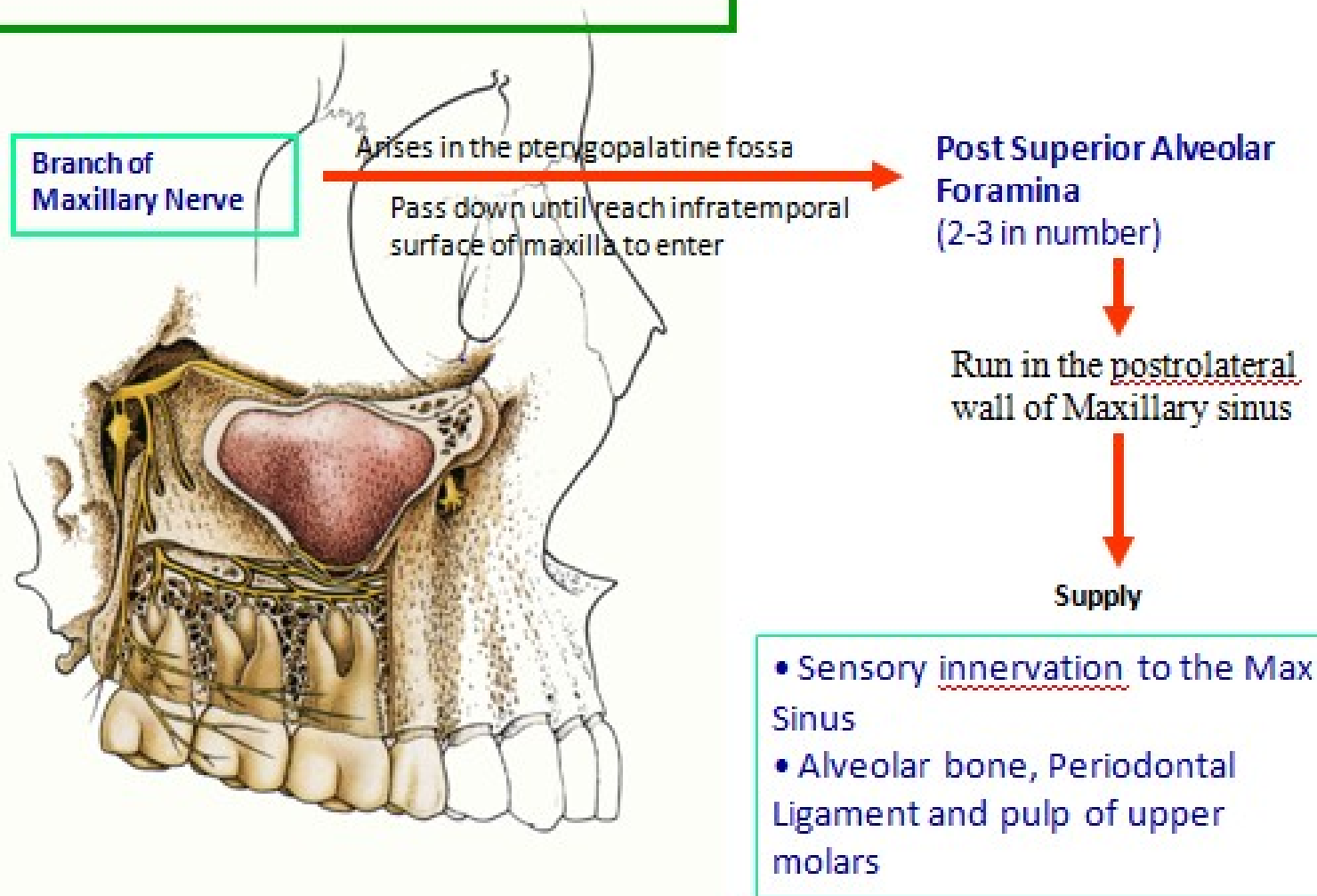
- Subjective: feeling of numbness in the area of administration
- Objective: use of electrical pulp testing (EPT) with no response from tooth with maximal EPT output (80/80)
- Absence of pain during treatment

Failures of anesthesia

- Needle tip below the apex
- Needle tip away from the bone

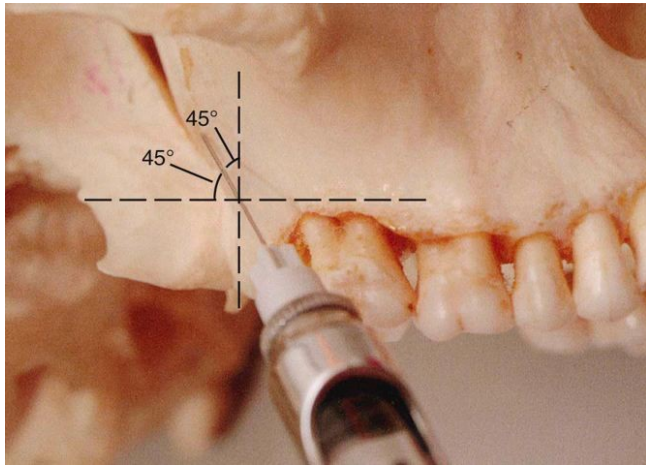


Posterior Superior Alveolar Nerve

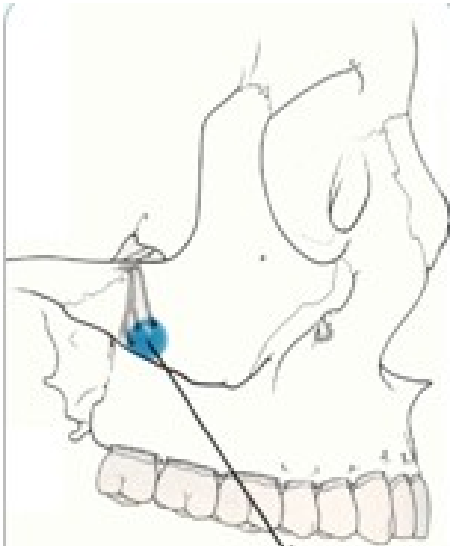


Posterior superior alveolar nerve block(PSA block)

- **Other names:** tuberosity block, zycomatic block
- **Areas anesthetized:** pulp of maxillary third,second and first molar (28% of patients the mesiobuccal root of first molar not anesthetized)
- Buccal periosteum and bone overlying these teeth



Infra temoral
surface of
maxilla



- **Indications**
 - When Supraperiosteal injection is contraindicated or two or more molar extractions are planned
- **Contraindications**
 - When possibility of hemorrhage is too risky, e.g.
 - Hemophiliac patients
 - Patients on Anticoagulant therapy
- **Disadvantages**
 - Risk of hematoma formation
 - Technique is too arbitrary as there is no bony landmarks

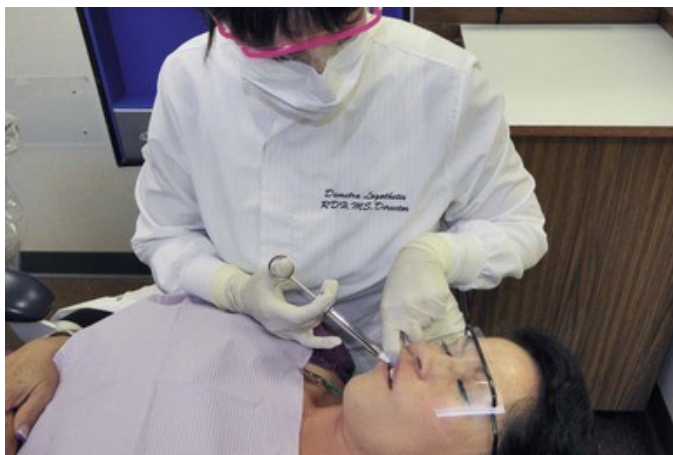
Technique

Short needle is used (20 mm)

Insertion at the mucobuccal fold above the maxillary second molar

Needle is advanced slowly in an upward, inward and posterior direction upto **16mm**

after negative aspiration , 1-1.5 ml is injected

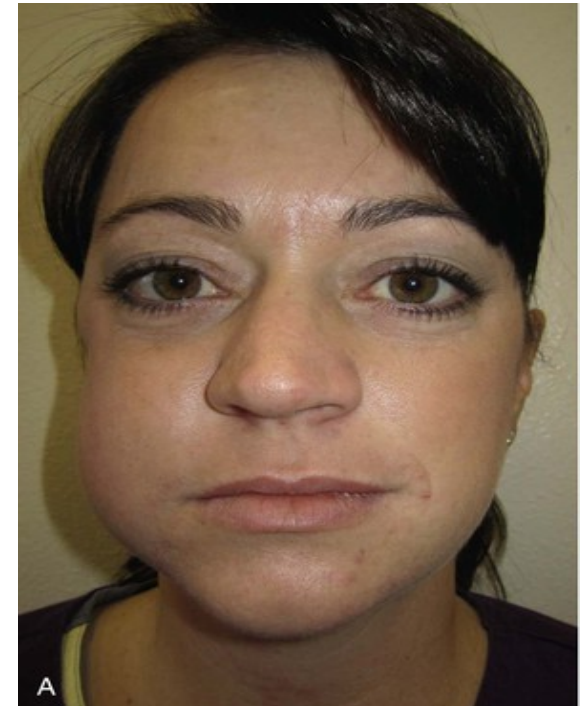
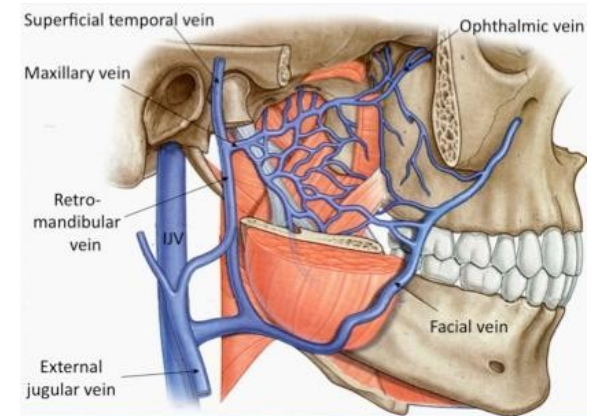


PSA Nerve block Complications

can occur if the needle is advanced **too far distally** into the tissue during a PSA block

The needle may penetrate the **pterygoid plexus of veins and the maxillary artery** if overinserted, possibly causing a **hematoma** in the infratemporal fossa.

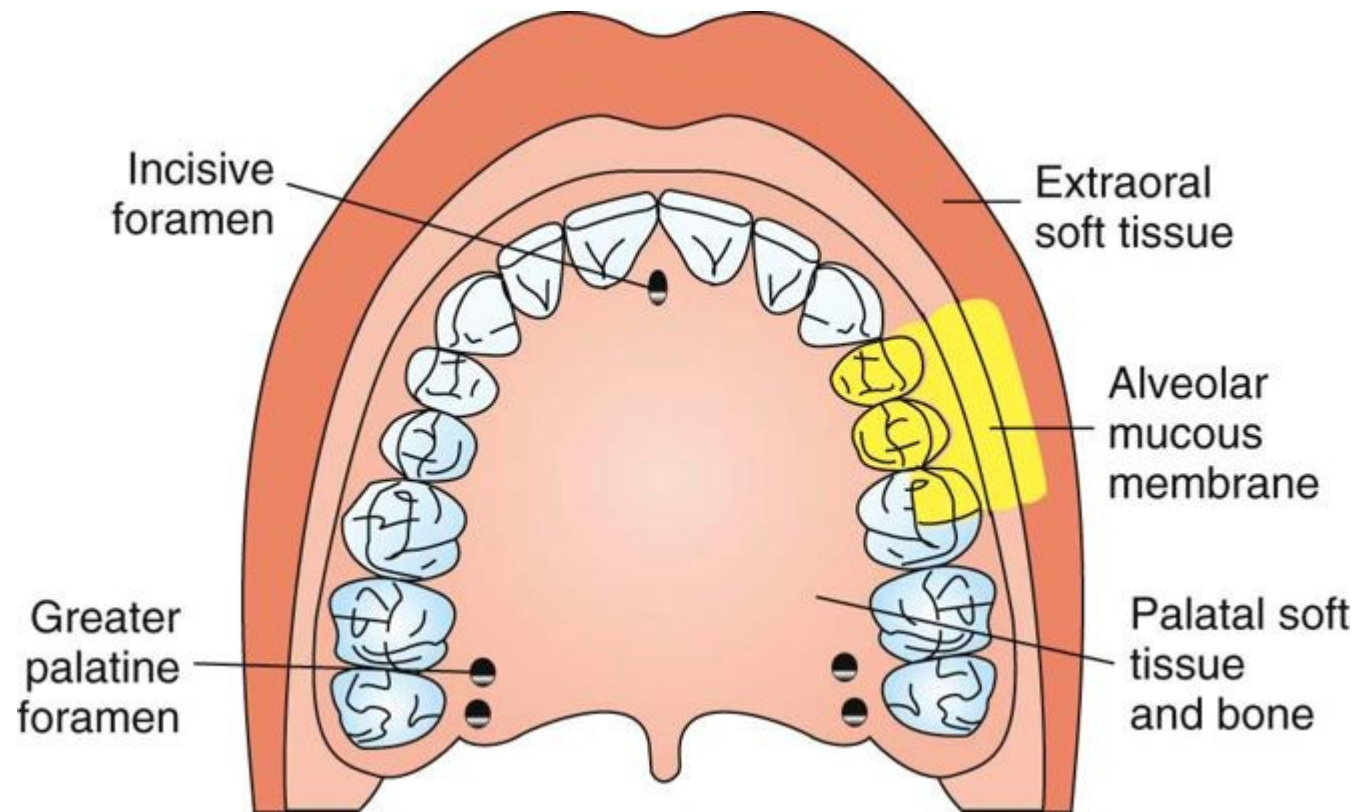
This results in a bluish-reddish extraoral swelling of hemorrhaging blood in the tissue on the affected side of the face in the infratemporal fossa a few minutes after the injection.



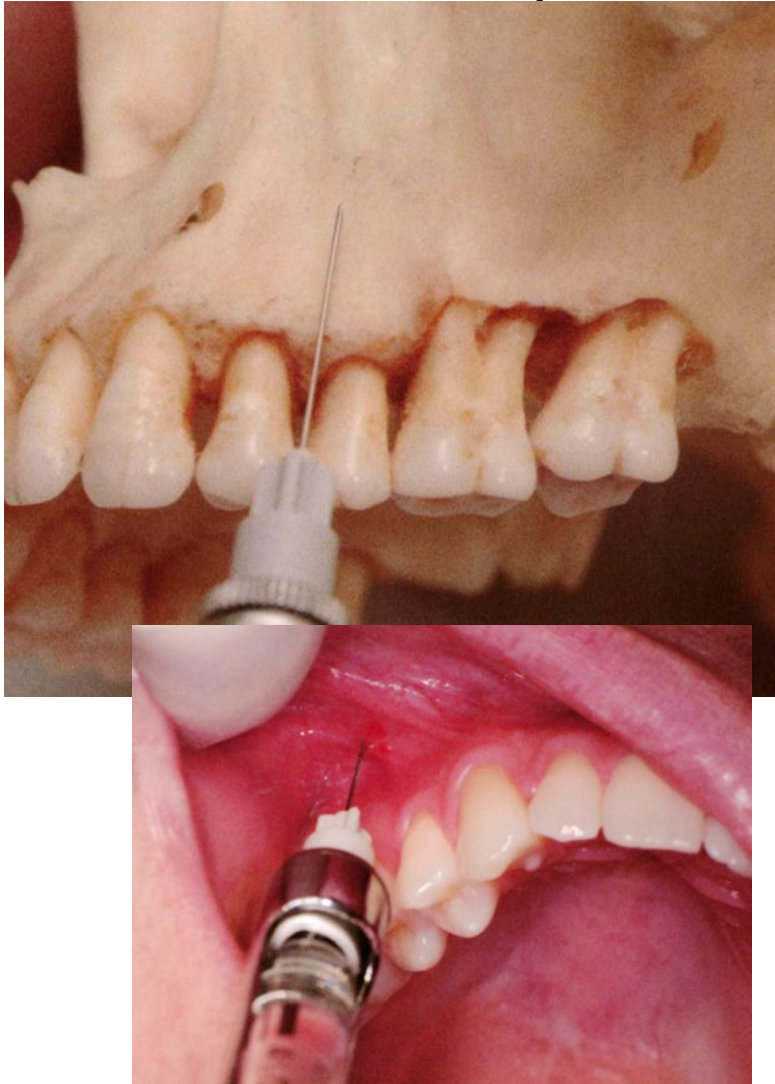
Middle Superior Alveolar Nerve Block

- The middle superior alveolar (MSA) nerve is present in only about 28% of the population.
- However, when the ASA nerve block fails to provide pulpal anesthesia distal to the maxillary canine, the MSA block is indicated for procedures on premolars and on the mesiobuccal root of the maxillary first molar. The success rate of the MSA nerve block is high.

Middle superior alveolar nerve block



Position of needle between maxillary premolars for a middle superior alveolar (MSA) nerve block.



Stretch the patient's upper lip to make the tissues taut .

Insert the needle into the height of the mucobuccal fold above the second premolar with the bevel directed toward bone.

Penetrate the mucous membrane and slowly advance the needle until its tip is located well above the apex of the second premolar

Aspirate & g Slowly deposit 0.9 to 1.2 mL (one half to two thirds cartridge) of solution

Withdraw the syringe and make the needle safe.

iWait a minimum of 3 to 5 minutes before commencing dental therapy.

(Infraorbital Nerve Block)

Local anesthetic injection into the infraorbital foramen, the anesthetic solution will diffuse through the infraorbital canal and anaesthetize the anterior and middle superior alveolar nerves

This is also called **infraorbital block**



Nerves Anesthetized:

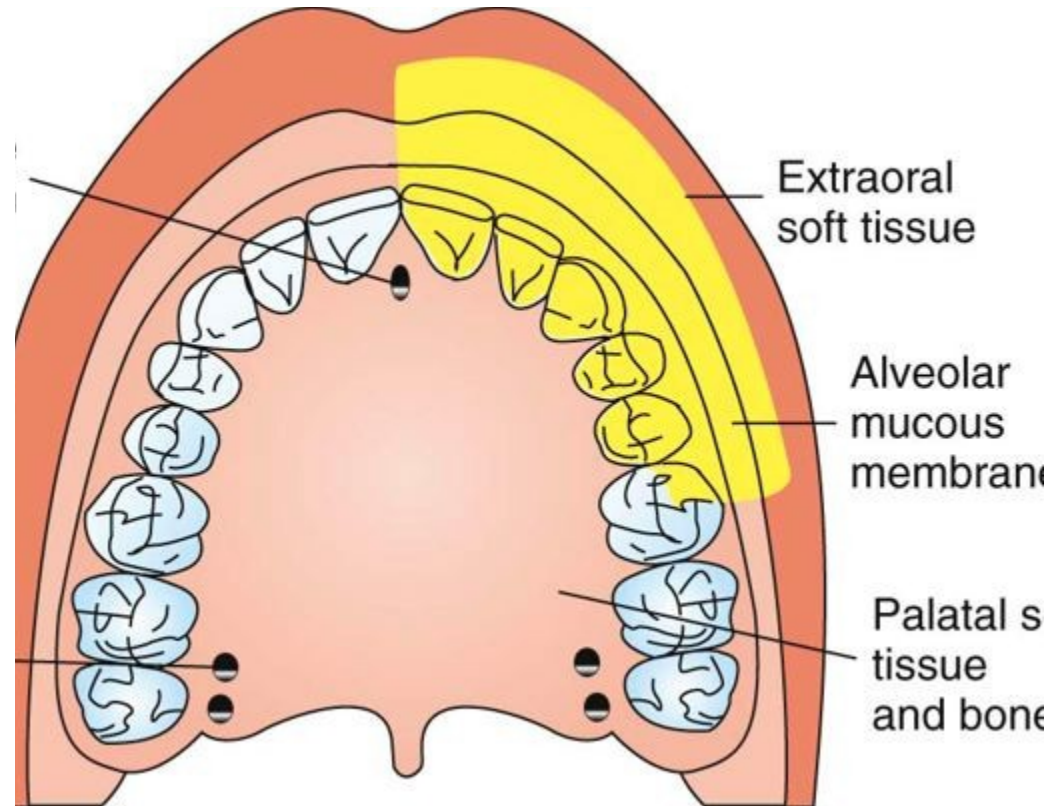
- Anterior Superior Alveolar nerve
- Middle Superior Alveolar nerve
- Infraorbital nerve branches
 - Inferior Palpebral Nerve
 - Lateral Nasal Nerve
 - Superior Labial Nerve

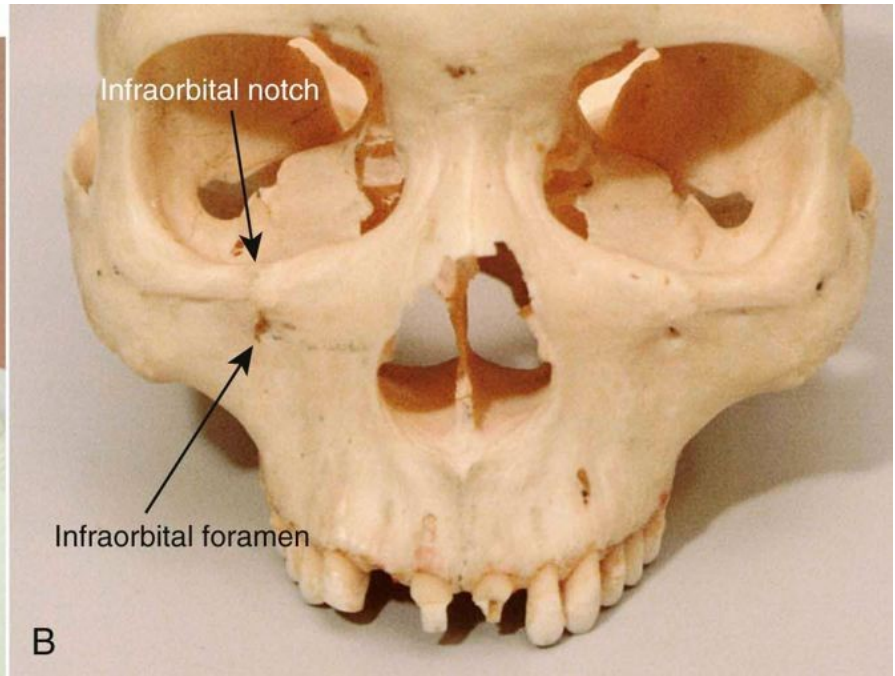
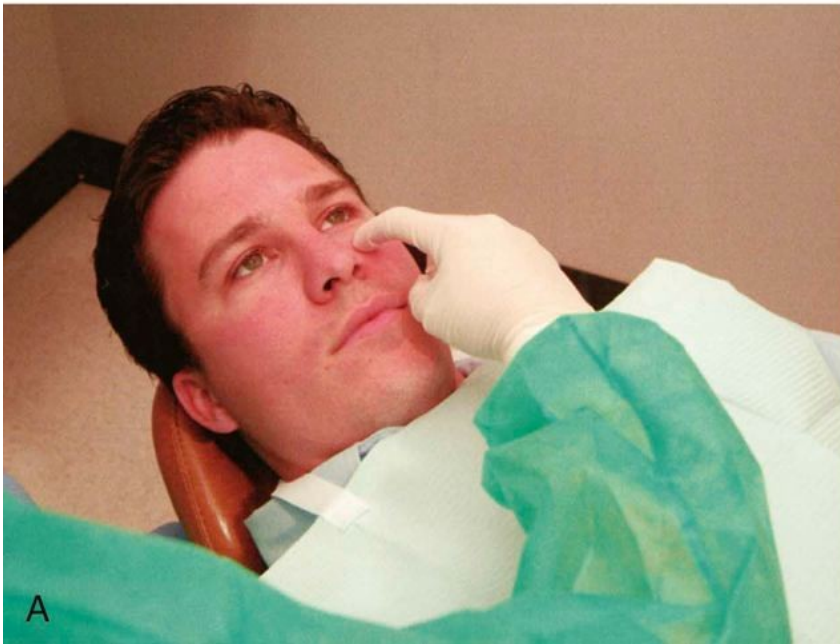
Areas Anesthetized:

Incisors Canine premolars
Mesiobuccal root of Upper first molar

Buccal periosteum and bone,
Soft tissues- Lower eyelid Portion
of the nose, Upper lip

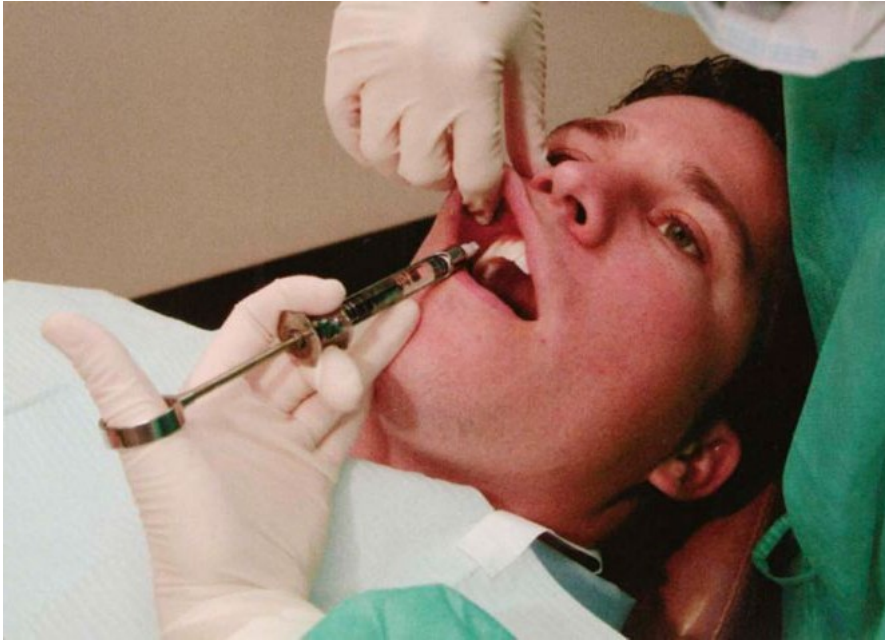
- **Anatomical Landmarks:**
mucobuccal fold,
Infraorbital notch, infra
orbital foramen





How to Locate the infraorbital foramen .

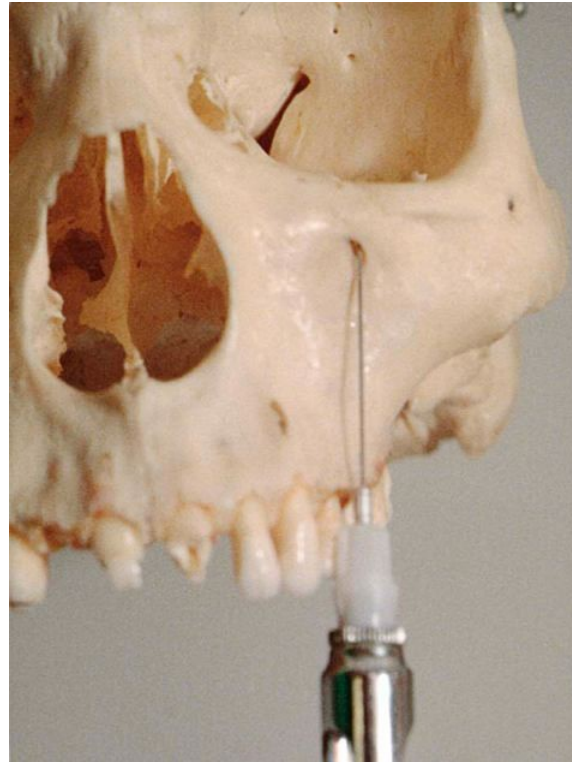
- (1) Feel the infraorbital notch.
- (2) Move your finger downward from the notch, applying gentle pressure to the tissues.
- (3) The bone immediately inferior to the notch is convex
- (4) As your finger continues inferiorly, a concavity is felt; this is the infraorbital foramen.
- (5) The patient senses a mild soreness when the foramen is palpated as the infraorbital nerve is pressed against bone.
- (6) Maintain your finger on the foramen or mark the skin at the site



Using a finger over the foramen, lift the lip, and hold the tissues in the mucobuccal fold taut

Insert the needle for anterior superior alveolar (ASA) nerve block in the mucobuccal fold over the maxillary first premolar.





Locating the Infraorbital Foramen

- Needle is placed parallel to the first premolar and slowly advanced to the depth
- Depth is about 16mm
- Slowly deposit 0.9 to 1.2 ml
- Maintain firm pressure with the finger to increase diffusion of LA into the infraorbital foramen

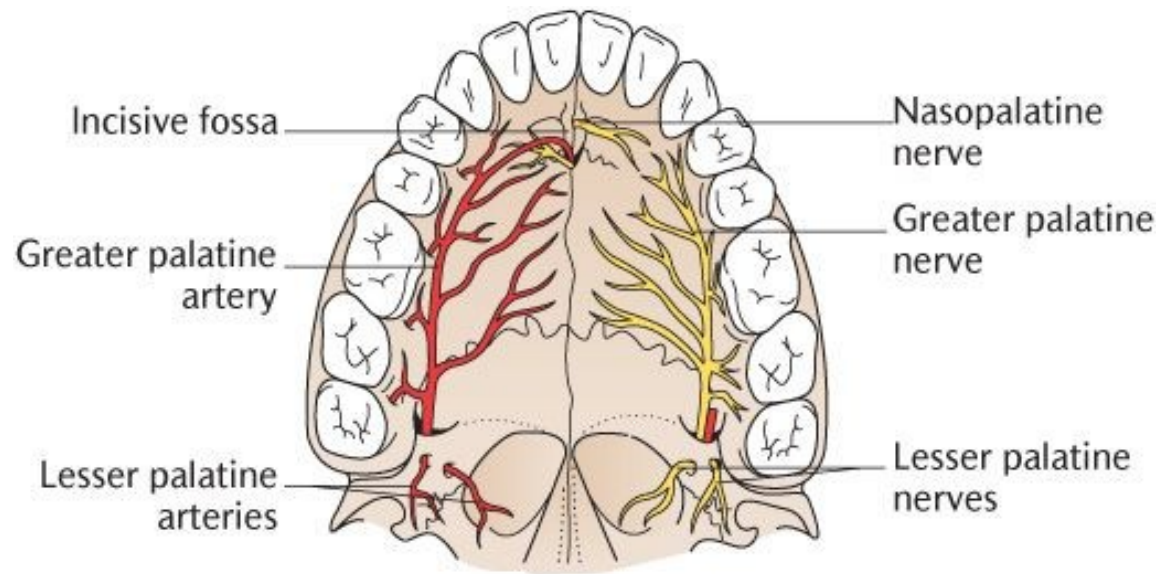
Failure of anesthesia

- Needle contacting below(inferior to foramen)
- Needle deviation to medial or lateral to the infraorbital foramen

Complications

hematoma -rare

Palatal anatomy



Palatal anesthesia

- For extraction and gingival surgery palatal anesthesia is essential to affect for hard and soft tissues.

Palatal anesthesia can be very painful

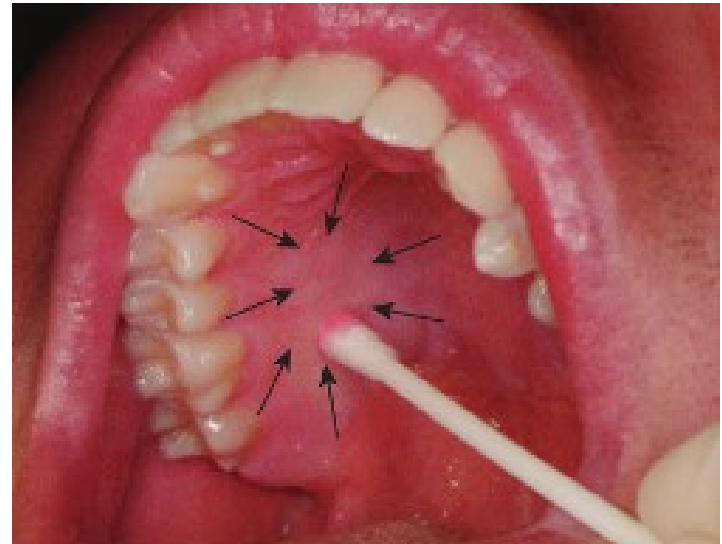
Steps in atraumatic palatal anesthesia

- apply topical anesthesia
- use pressure anesthesia
- maintain control over the needle
- deposit the solution very slowly



Pressure anesthesia

- Apply considerable pressure with a firm object
- Cotton applicator- better
- Pressure produces **ischemia** and **needle insertion is painless**



Slow injection

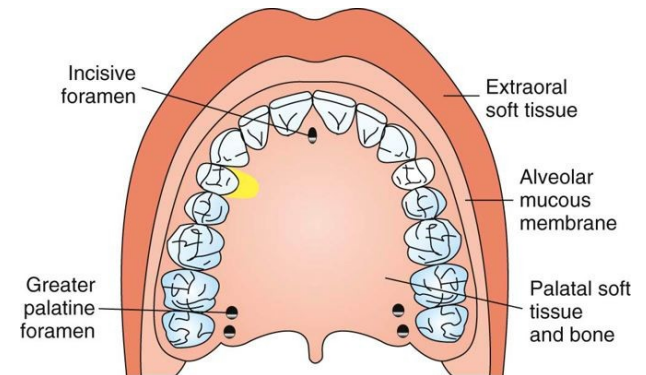
- Palatal soft tissues are dense
 - they are firmly adherent to bone
 - Rapid injection produces high tissue pressure, tear and lots of pain
- so slow injections are advised

Local infiltration of palate

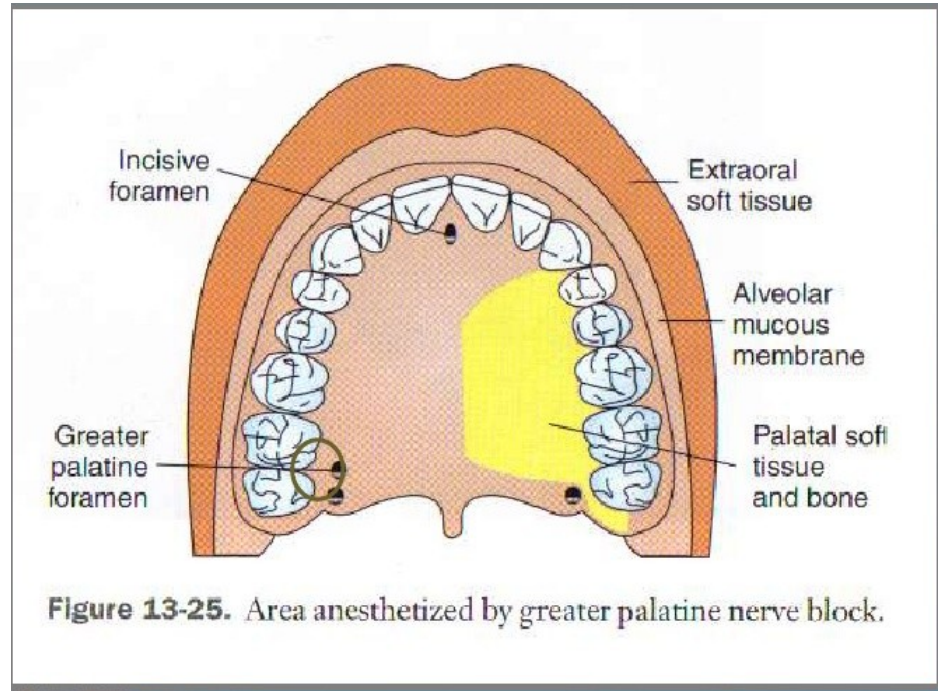
- Terminal branches of nasopalatine and greater palatine nerve anesthetized

The point of insertion is 5-10 mm away from the gingival margin

- The needle is directed from the other side to be at 45° angle
- **0.2 - 0.3 ml** of the solution is injected.
- Injection of large volume is painful and may cause sloughing and ulceration of the palatal mucosa



Greater Palatine Nerve Block

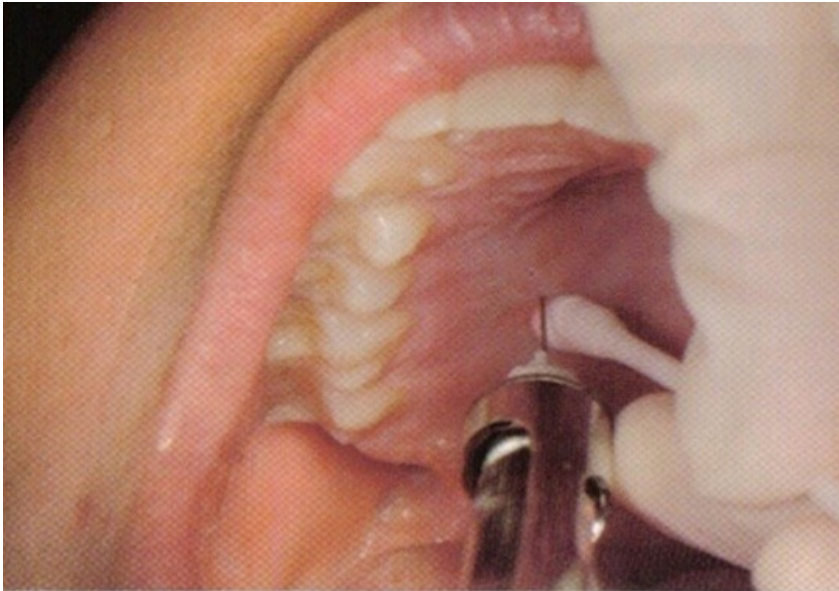


- The palatal gingiva, mucosa and periosteum of the molars and premolars are innervated with the **greater palatine nerve**
- The greater palatine foramen is located distal to the maxillary second molar on the palatal side



- The syringe is directed from the opposite side
- The foramen is located distal to maxillary second molar
- The needle is inserted into ischemic **tissues slightly anterior to the applicator stick.**
- the depth of penetration is about 5mm
- When the needle reaches bone, slowly inject **0.45-0.6ml**

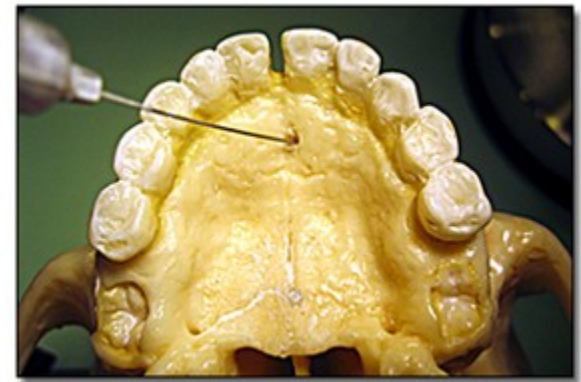
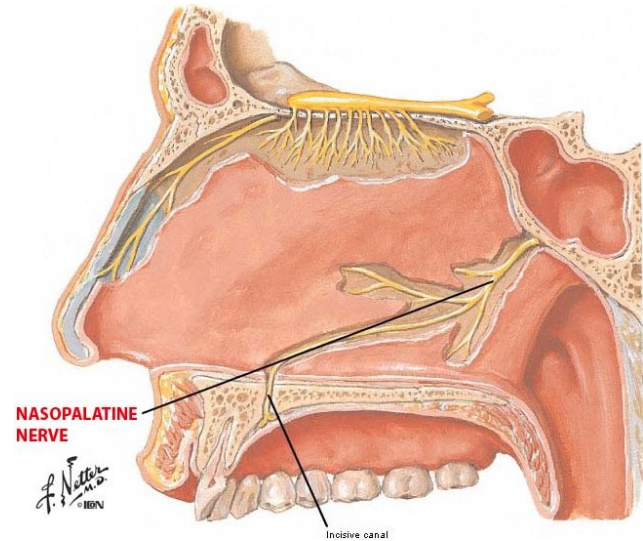
Failure of greater palatine nerve block



- If the local anesthetic is deposited too far anterior to the foramen, the palatal tissues posterior to the site of injection may not be anesthetized
- Inadequate Anesthesia on the palate, in the area of maxillary first premolar because of overlapping fibers from nasopalatine nerve

Nasopalatine nerve block

- Other names : incisive nerve, sphenopalatine nerve block
- Nerves anesthetized: nasopalatine nerves bilaterally
- Target : palatal tissues lateral to the incisive papilla

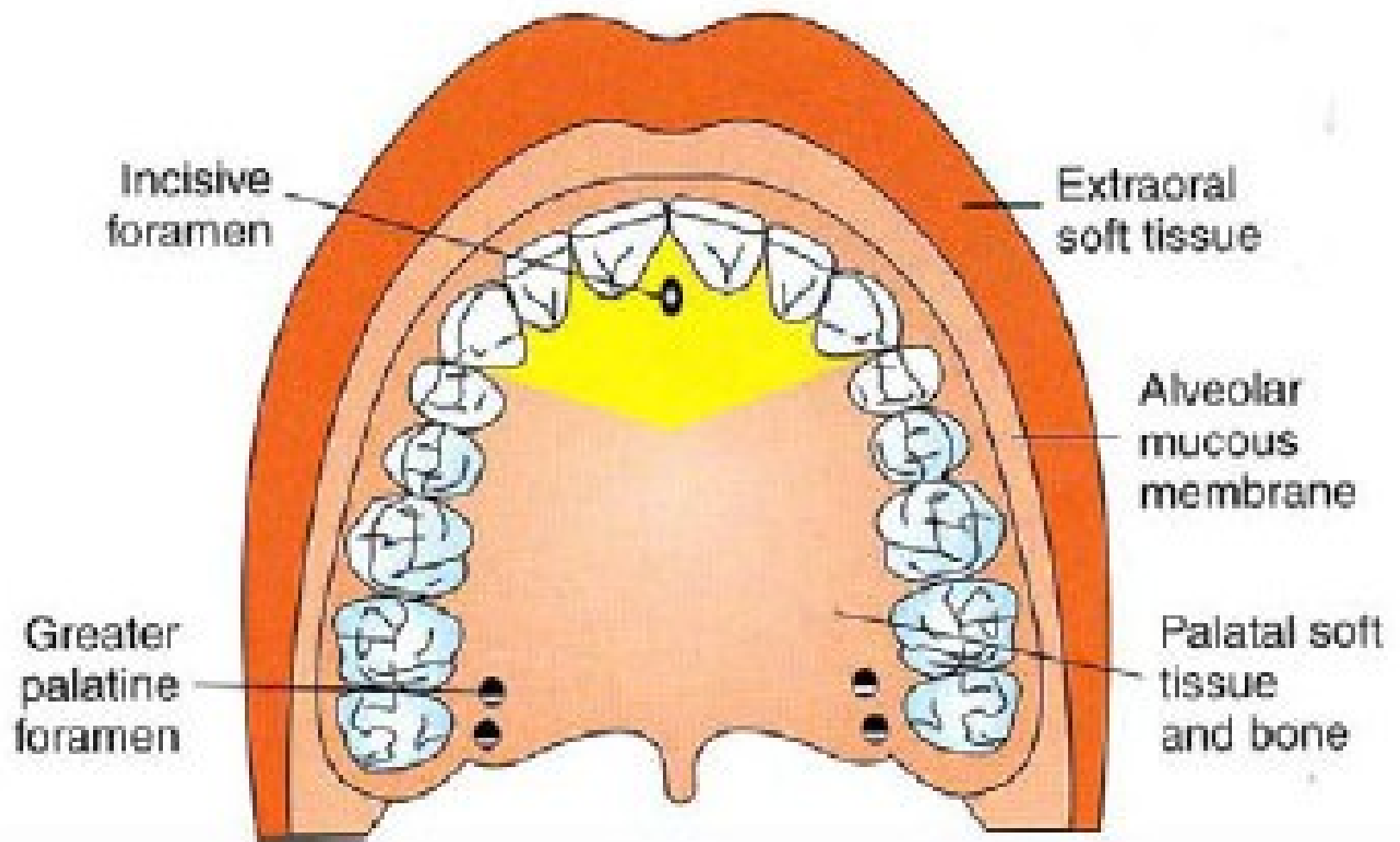


Procedure

- The Incisive papilla is very sensitive and injection into it is very painful
- so injection is made by insertion to lateral side of the papilla
- The needle is then advanced towards the incisive foramen
- The depth of penetration is not more than 5mm .
- Then **0.45 ml** of solution is deposited very slowly.



Figure 8. Distribution Of Anesthesia: Nasopalatine Nerve Block



Failures of anesthesia

- **Unilateral anesthesia**: if the solution is deposited to one side of canal, unilateral anesthesia can develop
- **Inadequate anesthesia** in the area of maxillary canine : because of overlapping fibers from greater palatine nerve

complications



- Hematoma- rare
- **Necrosis of soft tissue**- high concentration of vasoconstrictor
- The anesthetic solution may “squirt” back during injection- it is normal

Precautions:

Do not insert directly into incisive papilla – very painful

If the needle is inserted more than **5mm** ,may enter floor of the nose- infection may result

Some patients might be uncomfortable if their soft palate anesthetized

THE END